

Liam McKenna

work@liamdouglas.com | [\(321\) 607-2168](tel:(321)607-2168) | liamdouglas.com | [LinkedIn](#) | [GitHub](#)

TECHNICAL SKILLS

Languages: C++, C#, GLSL, Lua, Java, Batch

Tools & Frameworks: Unreal Engine, Unity, OpenGL, DirectX, Blender, Maya, Visual Studio

Specialized Knowledge: Graphics Programming, Game Engine Architecture Design, 3D Modeling ([Portfolio](#))

EDUCATION

Master of Science (M.S.), Computer Science

University of Florida, Kyoto University

- 3.90 GPA

Spring 2026

Gainesville, FL

Bachelor of Science (B.S.), Computer Science

University of Florida

- Cum Laude (3.58 GPA)

Summer 2024

Gainesville, FL

EXPERIENCE

Academic Researcher

Kyoto University — Project AirSim

Aug 2025 - Dec 2025

Kyoto, JP

- Studied and worked within Unreal Engine's C++ source code to develop a fork used for computer vision research
- Integrated new menu systems into existing codebase to greatly increase program automation and accessibility
- Contributed significant production-grade code to a popular open-source repository with an active community

Academic Researcher

University of Florida SurfLab

Jan 2025 - Aug 2025

Gainesville, FL

- Investigated experimental approaches to producing global illumination in a real-time OpenGL environment
- Attained performant pixel-accurate shadow casting of complex NURBS surfaces with novel use of past research
- Developed efficient two-way GPU data transfer tools using GLSL and C++ for resource and performance analysis

Software Development Intern

United Wholesale Mortgage

May 2023 - Aug 2023

Pontiac, MI

- Garnered applied experience in the agile development cycle by operating under the scrum framework
- Developed an integrated software stability inspector with C# for end-to-end use in proprietary software

PROJECTS

PowerLine | Original Minecraft Multiplayer Minigame

Java

Aug 2025 - Sep 2025

[GitHub](#) | [Play Game](#)

- Led the full development cycle of a novel multiplayer game mode for Minecraft through custom plugin programming
- Wrote ~3,800 LOC over a two-week sprint to fully implement the gameplay systems required for the project
- Achieved publication on the popular and selective Minecraft minigame hosting platform, StickyPiston

APGP | Multipurpose Custom 3D Rendering Environment

C++, OpenGL, GLSL, Lua

Jul 2024 - Aug 2025

[GitHub](#)

- Created a highly modular real-time 3D application in C++ using the OpenGL graphics API
- Implemented an Entity-Component System (ECS), Scene Graph, and support for Physically-Based Rendering (PBR)
- Engineered fully dynamic Lua script insertion, asset management, and scene generation at runtime

Topposition | Procedurally Generated Game Built on Custom 2D Engine

C++, SFML

Aug 2023 - Dec 2023

[GitHub](#) | [Presentation](#)

- Constructed a 2D game engine in C++ from the ground up only using SFML for rendering
- Developed a feature-complete strategy game to utilize the engine featuring a procedurally generated terrain system
- Presented a lecture to UF's game development club, DevLUp, on the lessons learned throughout development

Itch.io Showcase | Extended Game Development Portfolio

Unity, C#, Batch

Aug 2022 - May 2023

[Portfolio](#)

- Attained extensive experience working in Unity and C# scripting from the development of several unique projects
- Achieved multiple top placements and academic recognition in game jams, hackathons, and class projects